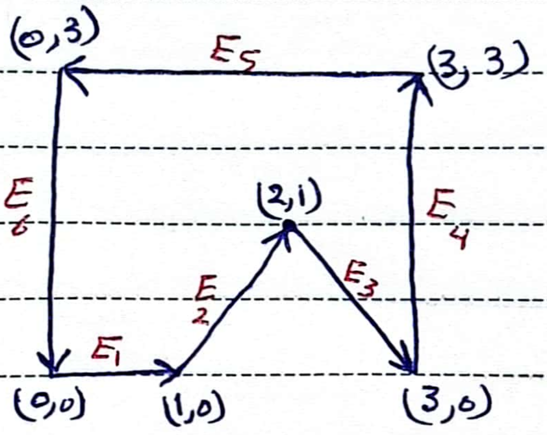
Sec 8 : filling

* to find the vector between 2 points

$E_1 \rightarrow (X_2 - X_1 , y_2 - y_1 , Z_2 - Z_1)$

Ex 1) find the vectors & check if it's ConCave polygon



$E_1 = (1-0 , 0-0 , 0) = (1,0,0)$

$= (\boxed{1},\boxed{0},0)$

$E_1 X E_2 = (0,0, 1*1 - 0*1)$
$= (0,0,1)$

$E_2 = (\boxed{1},\boxed{1},0)$

$E_2 X E_3 = (0,0, 1*-1 - 1*1)$
$= (0,0,\boxed{-2})$ ∴ ConCave

$E_3 = (1,-1,0)$

$E_3 X E_4 = (0,0, 1*3 - 0*-1)$
$= (0,0,3)$

$E_4 = (0,3,0)$

$E_4 X E_5 = (0,0, 0-(-9))$
$(0,0,9)$

$E_5 = (-3,0,0)$

$E_5 X E_6 = (0,0,9)$

$E_6 = (0,-3,0)$

$E_6 X E_1 = (0,0, 0*0 - (-3*1))$
$= (0,0,3)$

Note: * the Cross product bet 2 vectors (X,y) represent (Z) vector

* to check if ConCave

$E_i X E_j < 0$ for any vector in the polygon

Filling algorithms :

1 Boundary fill : (edge fill)

① Start at a point inside the polygon boundary

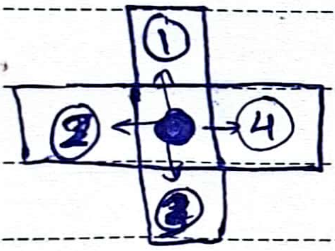
② Check on the neighboring pixel if → not Colored & have not boundry Color → then Color it

* only work if filled & boundry Color are different

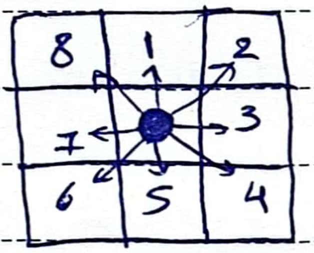
③ Store its neighboring pixels 4N or 8N in the stack

④ Select next pixel from stack & repeat ②

4 Neighbors



8 N

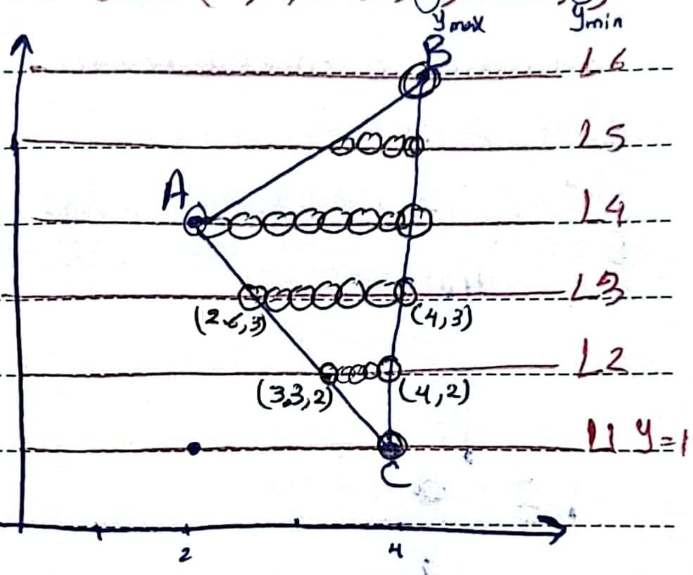


2 flood fill (seed fill)

if you want to recolor the area it replace the interior Color with fill Color & start from an interior point (seed point)

3 Scan line polygon fill :

Ex 1 Using scanline polygon fill algorithm to determine the content of the active edge table to fill the polygon with vertices A(2,4) B(4,6) C(4,1) for y = 1 to 6



① ارتب ال edges من اقل y لأعلى"
كانه الترتيب من الشكل ازاي :

∴ C(4,1) → A(2,4) → B(4,6)

③ هحط ال y من ال min لل max

② عندكل y هحط ال vertex اللى بتقاطع معاها
لو مفيش احط Ø

y	vertex	
max 6	B	→ null
5	Ø	→ null
4	A	AB → 6 | 2 | 1
3	Ø	→ null
2	Ø	→ null
min 1	C	CA → 4 | 4 | -2/3 → CB 6 | 4 | 0

④ اعمل ال global edge table (GET)
(array of linked lists)

* CA → $(\overset{x_1}{4},\overset{y_1}{1})$ to $(\overset{x_2}{2},\overset{y_2}{4})$

$y_{max} = 4$	$X_{y_{min}} = 4$	$\frac{1}{m} = \frac{-2}{3}$

$y_2 > y_1$

x_1 for y_1 is min

$$\frac{\Delta X}{\Delta y} = \frac{2-4}{4-1} = \boxed{\frac{-2}{3}}$$

* CB → (4,1) to (4,6)
max

6	4	0

$$\frac{1}{m} = \frac{4-4}{6-1} = \frac{0}{5} = 0$$

⑤ من اخر واحد همشي عند كل vertex واشوف شروطه بأنهي edge واعمل ال linked list بتاعتها والمتكرر مش نكتبه

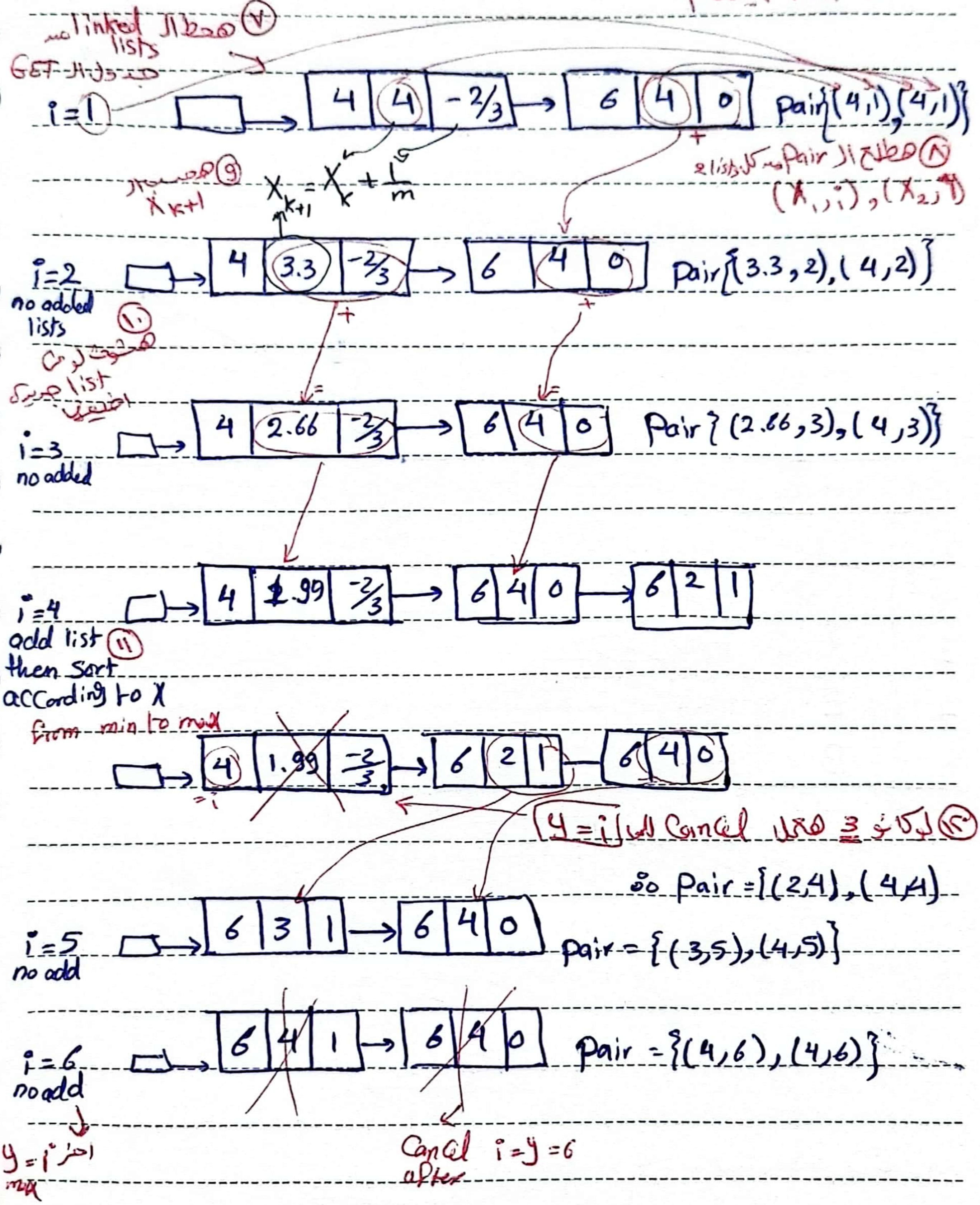
* AB → (2,4) to (4,6)

6	2	1

$$\frac{1}{m} = \frac{4-2}{6-4} = \frac{2}{2} = 1$$

⑥ هحسب ال active edge table عند كل i تتمثل y بطلع ال pair الى point

هعمل fill جديد


i=1
4 | 4 | -2/3
6 | 4 | 0
+
i=2
4 | 3.3 | -2/3
6 | 4 | 0
+
+
i=3
4 | 2.66 | -2/3
6 | 4 | 0
i=4
4 | 1.99 | -2/3
6 | 4 | 0
6 | 2 | 1
4 | 1.99 | -2/3
6 | 2 | 1
6 | 4 | 0
=i
i=5
6 | 3 | 1
6 | 4 | 0
i=6
6 | 4 | 1
6 | 4 | 0

⑦ هحط ال linked lists من GET ال ده

i=1: Pair{(4,1),(4,1)}

⑧ هطلع ال pair من كل 2 lists

$(X_1, i), (X_2, i)$

⑨ هحسب ال X_{k+1}

$$X_{k+1} = X_k + \frac{1}{m}$$

i=2 no added lists: Pair{(3.3,2),(4,2)}

⑩ هشوف لو في list جديد اضيفه

i=3 no added: Pair{(2.66,3),(4,3)}

i=4 add list ⑪ then sort according to X from min to max

⑫ لو كان في 3 هعمل ال cancel لل y=i

so Pair={(2,4),(4,4)}

i=5 no add: pair={(3,5),(4,5)}

i=6 no add: pair={(4,6),(4,6)}

$y = i$ احزر max

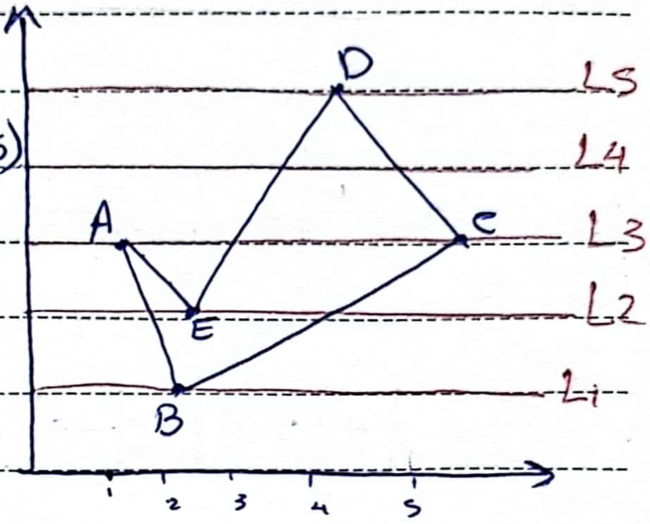
Cancel after i=y=6

DAWENHA designs

Ex. 2 Using Scanline fill to determine the Content of the active edge table to fill the polygon with vertices A(1,3), B(2,$\underset{\downarrow y_{min}}{1}$), C(5,3), D(4,$\underset{\downarrow y_{max}}{5}$) E(2,2)
y from 1 to 5

① Sort vertex:

B(2,1), E(2,2), A(1,3), C(5,3), D(4,5)



② Creat GET

y	Vertex	
5	D	→ null
4	0	→ null
3	A, C	$\xrightarrow{CD}\boxed{5}\boxed{5}\boxed{-1/2}$
2	E	$\xrightarrow{EA}\boxed{3}\boxed{2}\boxed{-1}\xrightarrow{ED}\boxed{5}\boxed{2}\boxed{2/3}$
1	B	$\xrightarrow{BA}\boxed{3}\boxed{2}\boxed{-1/2}\xrightarrow{BC}\boxed{3}\boxed{2}\boxed{3/2}$

* for BA → (2,1)(1,3) $\boxed{3}\boxed{2}\boxed{-1/2}$ $\frac{1-2}{3-1} = \frac{-1}{2}$

for BC → (2,1)(5,3) $\boxed{3}\boxed{2}\boxed{3/2}$ $\frac{5-2}{3-1} = \frac{3}{2}$

for EA → (2,2)(1,3) $\boxed{3}\boxed{2}\boxed{-1}$ $\frac{1-2}{3-2} = -1$

for ED → (2,2)(4,5) $\boxed{5}\boxed{2}\boxed{2/3}$ $\frac{4-2}{5-2} = \frac{2}{3}$

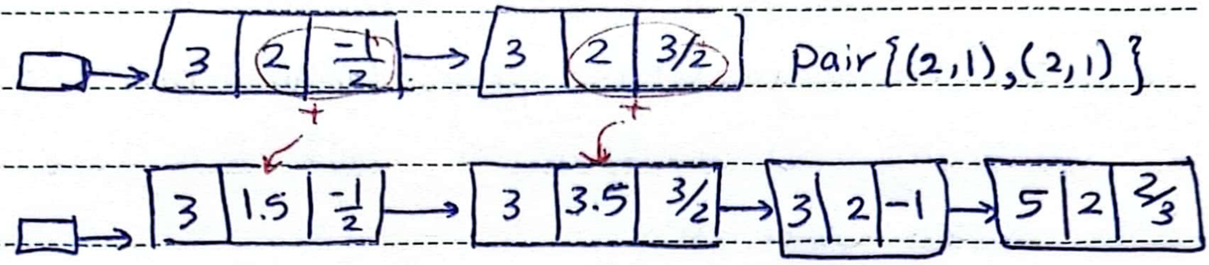
for CD → (5,3)(4,5) $\boxed{5}\boxed{5}\boxed{-1/2}$ $\frac{4-5}{5-3} = \frac{-1}{2}$

③ Create active edge table :-

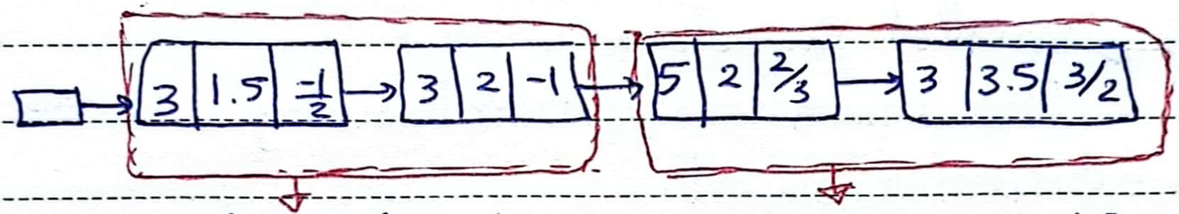
i=1

pair{(2,1),(2,1)}

i=2 added


3
2
$\frac{-1}{2}$
3
2
3/2
+
+
3
1.5
$\frac{-1}{2}$
3
3.5
3/2
3
2
-1
5
2
2/3

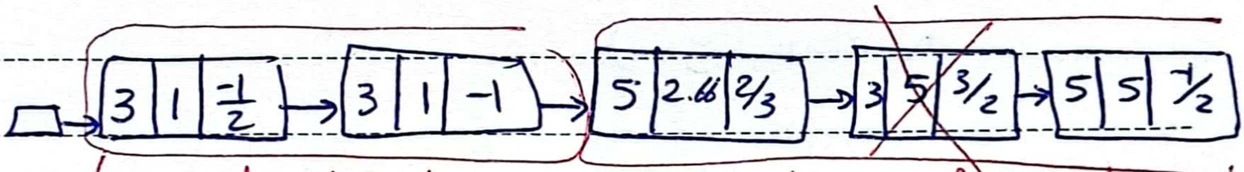
Sort according to X from min to max


3
1.5
$\frac{-1}{2}$
3
2
-1
5
2
2/3
3
3.5
3/2

So Pairs{(1.5,2)(2,2)}

pair={(2,2)(3.5,2)}

i=3 added


3
1
$\frac{-1}{2}$
3
1
-1
5
2.66
2/3
3
~~5~~
~~3/2~~
5
5
-1/2

Sorted already then take each pair of lists

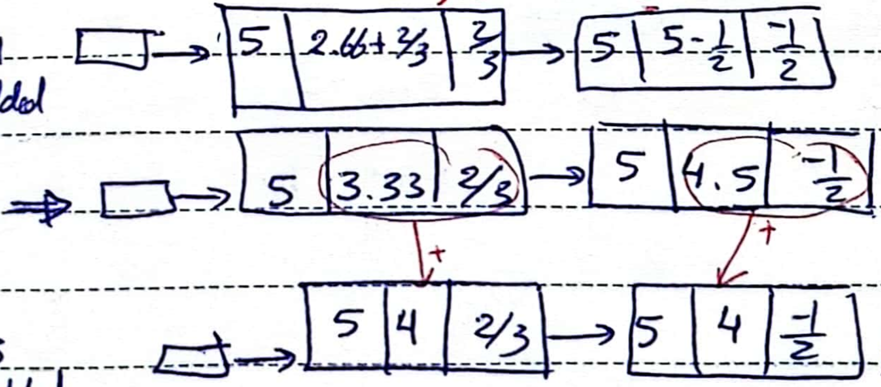
pair={(-1,3)(-1,3)}

then Cancel when i=4 because its greater than y

Cancel y=i then get pair

pair={(2.66,3)(5,3)}

i=4 no added


5
2.66+2/3
2/3
5
$5-\frac{1}{2}$
$\frac{-1}{2}$
5
3.33
2/3
5
4.5
$\frac{-1}{2}$
+
+
5
4
2/3
5
4
$\frac{-1}{2}$

pair{(3.33,4)(4.5,4)}

i=5 no added

pair{(4,5),(4,5)}

last $i=y_{max}=5$

Sec 9: projection

refer to the mathimatical process (matrix multiplication) of converting 3D Coordinates of points in a scene into screen Coordinates

1 → parallel projection
(by drawing straight lines from every point on the object

2 → prespective projection
(represent objects as perceived by the human eyes)

* Prespective projection matrix

field of view angle

aspect ratio

d: field of view scaling factor

$$\begin{bmatrix} x' \\ y' \\ z' \\ w \end{bmatrix} = \begin{bmatrix} \cot\left(\frac{fov}{2}\right) * \frac{1}{a} & 0 & 0 & 0 \\ 0 & \boxed{\cot\left(\frac{fov}{2}\right)} & 0 & 0 \\ 0 & 0 & \boxed{\frac{Z_{near}+Z_{far}}{Z_{near}-Z_{far}}} & \boxed{\frac{-2Z_{near}Z_{far}}{Z_{near}-Z_{far}}} \\ 0 & 0 & -1 & 0 \end{bmatrix} * \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix}$$

S_z scaling factor

t_z transfer factor

Ex.1 project the point (2,3,5) onto a screen using a camera with a fov of 90°, aspect ratio of 16:9 & near/far planes at 1 and 100 units

z_{near} z_{far}

$$a = \frac{16}{9} = 1.77, \quad d = \cot\left(\frac{90}{2} * \overset{3.14}{\pi}/180\right) = 1, \quad S = \frac{1+100}{1-100} = -1.02$$

$$t = \frac{-2 * 1 * 100}{1-100} = 2.02$$

$$\begin{bmatrix} x' \\ y' \\ z' \\ w \end{bmatrix} = \begin{bmatrix} d/a & 0 & 0 & 0 \\ 0 & d & 0 & 0 \\ 0 & 0 & S & -t \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ 5 \\ 1 \end{bmatrix}$$

$x' = \frac{2d}{a}$

$y' = 3d$

$z' = 5S - t$

* prefor perspective divide :

$x' = \frac{x'}{z} = \frac{}{5}$ $\quad y' = \frac{y'}{5}$ $\quad z' = \frac{z'}{5}$